

Supplementary COS- Ξ Bridge Results

Dmitry Rakitin
Independent Researcher
phizmat17@gmail.com

14 September 2026

These are supplementary paper-level bridge statements. They do not have the same independent-audit status as the main theorem suite.

B1: finite-window critical-line transfer

Fix a finite ordinate window $[T, 2T]$ and retained simple critical-line zeros

$$\Gamma_T = \{\gamma_1 < \cdots < \gamma_s\}, \quad \Xi(\gamma_j) = 0, \quad \Xi'(\gamma_j) \neq 0.$$

For any prescribed $\eta_T > 0$ and lower threshold $\Lambda_*(T)$ there is $\Lambda_{C1}(T) \geq \Lambda_*(T)$ such that every $\lambda \geq \Lambda_{C1}(T)$, with

$$t = \frac{e^\lambda}{1 + e^\lambda},$$

gives for each j a unique simple real zero $r_j(t)$ of $H_t = (1 - t) \cos + t\Xi$ satisfying

$$|r_j(t) - \gamma_j| < \eta_T.$$

For disjoint localization neighborhoods the order is preserved.

B2: local multiplicity splitting

Let f have a real zero γ of multiplicity $m \geq 1$, and let $g(\gamma) \neq 0$. For all sufficiently small real $\varepsilon \neq 0$, each perturbation $f + \varepsilon g$ and $f - \varepsilon g$ has exactly m simple roots in a sufficiently small disk around γ . Across the two perturbations together, exactly two roots are real and

$$\boxed{2m - 2}$$

are nonreal.

In the Ξ application one may choose a cosine direction nonvanishing at the retained zero; the campaign uses the pair $\cos z$ and $\cos(\sqrt{2}z)$ to avoid simultaneous vanishing.

B3: finite-window defect identities

Let

$$E_{\mathbb{R}} = \sum_{\gamma \text{ real}} (m_{\gamma} - 1)$$

over the retained real zeros in pairwise disjoint localization disks, and let $D_{\mathbb{R}}$ count the paired local nonreal roots generated from them. Then

$$\boxed{D_{\mathbb{R}} = 2E_{\mathbb{R}}}.$$

If N_{off} is the total multiplicity of retained nonreal Ξ -zeros in the same finite localization scheme, then the full paired local defect obeys

$$\boxed{D_{\text{all}} = 2N_{\text{off}} + 2E_{\mathbb{R}}}.$$

These are finite/local identities. No endpoint-scale $o(N)$ global defect budget is claimed.

Status

B1 is recorded as `C1 = SOLVED_AT_PAPER_LEVEL`. B2 and B3 are paper-level local bridge results. They remain separate from the independently audited main COS- Ξ theorem suite.